PORTFOLIO DURATION AND CONVEXITY

Learning Outcome

- calculate portfolio duration and convexity and explain the limitations of these measures

Just as duration and convexity provide measures of the interest rate risk of a single bond, these metrics can also be used to measure the interest rate risk of a portfolio of bonds. There are two ways to calculate the duration and convexity of a bond portfolio:

1. Using the weighted average of time to receipt of the *aggregate* cash flows
2. Using the weighted averages of the durations and convexities of the individual bonds that make up the portfolio

The first method is the theoretically correct approach, but it is difficult to use in practice. Our focus will be the second method since it is commonly used by fixed-income portfolio managers. We will also highlight its limitations.

Suppose an institutional investor holds a two-bond portfolio consisting of $50 million par value each of the BRWA five-year bond and a government of Romania 30-year bond, with details as shown in Exhibit 6. Assume that both bonds are denominated in US dollars.

Exhibit 6:

Inputs for Determining Portfolio Duration and Convexity

Bond	Maturity (Years)	Coupon (%)	Price	Yield (%)	Duration	Convexity
BRWA	5	3.200	100.000	3.200	4.58676	24.23896
Romania	30	4.625	98.022	4.750	15.90637	369.64203

Bond	Par Value	Market Value	Portfolio Weight
BRWA	50,000,000	50,000,000	0.5050
Romanian	50,000,000	49,011,224	0.4950

The total market value for the portfolio is $99,011,224, and the portfolio is nearly evenly weighted in terms of market value between the two bonds. As noted previously, duration and convexity for a portfolio are calculated as the weighted average of the statistics for the individual bonds, with the shares of overall portfolio market value as the weights. This weighted average approximates the theoretically correct portfolio duration and convexity and becomes more accurate when differences in the yields-to-maturity on the bonds are smaller and when the yield curve is flat.

Given the portfolio weights in this simple example, portfolio duration and convexity are calculated as follows:

Weighted-average modified duration = (4.58676 × 0.5050) + (15.90637 × 0.4950) = 10.19004.

Weighted-average convexity = (24.23896 × 0.5050) + (369.64203 × 0.4950) = 195.21581.

The main advantage of this weighted-average approach is that it is easily used as a measure of interest rate risk. For instance, if yields-to-maturity on each of the bonds in the portfolio increase by 100 bps, then the estimated decline in the portfolio value is 9.214%.

and convexity implicitly assume that yields of all maturities change by the same amount in the same direction, a **parallel shift** in the yield curve. In reality, pure parallel shifts are rare; rather, we frequently observe a steepening, flattening, or even twisting yield curve.

EXAMPLE 2

Portfolio Duration and Convexity

An institutional investor considers adding a new USD50 million par value position in a 10-year US Treasury bond to its existing portfolio of BRWA and government of Romania bonds. The relevant data are shown in Exhibit 7.

Exhibit 7:

Inputs for Determining Portfolio Duration and Convexity

Bond	Maturity (Years)	Coupon (%)	Price	Yield (%)	Duration	Convexity
BRWA	5	3.200	100.000	3.200	4.58676	24.23896
Romanian	30	4.625	98.022	4.750	15.90637	369.64203
US Treasury	10	1.500	98.168	1.700	9.23693	93.87376

Bond	Par Value (USD)	Market Value (USD)	Portfolio Weight
BRWA	50,000,000	50,000,000	0.33762
Romania	50,000,000	49,011,224	0.33094
US Treasury	50,000,000	49,083,948	0.33144

1. Calculate the weighted-average duration and convexity for the proposed portfolio.

   Hide Solution

   Solution:

   Given the data in Exhibit 7, the weighted-average modified duration and convexity are calculated as follows:

   Weighted-average modified duration = (4.58676 × 0.33762) + (15.90637 × 0.33094) + (9.23693 × 0.33144) = 9.87415.

   Weighted-average convexity = (24.23896 × 0.33762) + (369.64203 × 0.33094) + (93.87376 × 0.33144) = 161.62749.

2. Compare and interpret duration and convexity for the proposed portfolio versus the current portfolio.

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   Solution:

   Adding the US Treasury position would *decrease* both the portfolio duration (from 10.19004 to 9.87415) and convexity (from 195.21581 to 161.62749). The reduction in duration would reduce the price risk of the portfolio against an upward parallel shift in the yield curve, but due to the lower convexity, this reduction in risk would be lessened for large shifts.

3. Recommend whether the US Treasury bond position should be added if the investor expects a 100 bp parallel shift downward in yields.

Given an expected 100 bp parallel shift down in yields, the investor should not add the position in US Treasury bonds. Adding the position lowers both the portfolio duration and convexity, which would also reduce the expected increase in the value of the portfolio. Given the investor's yield curve view, it should seek to increase both portfolio duration and convexity.

Question Set

Questions 1–4 relate to the following information.

An investor purchases EUR10 million par value of a 5-year, zero-coupon bond and a 10-year, fixed-rate semiannual coupon bond. Details of the bonds are shown below.

Two-Bond Portfolio

Bond	Maturity (Years)	Coupon (%)	Price	YTM (%)	Duration	Convexity
Zero	5	0.00	83.18777	3.750	4.81928	27.87052
Semi-annual	10	5.50	105.91556	4.750	7.71210	72.54897

1. Calculate the weighted-average modified duration for the portfolio.

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Solution:

The market value of the bonds are as follows:

Zero-coupon: €10,000,000 × 83.18777/100 = €8,318,777.

Semi-annual: €10,000,000 × 105.91556/100 = €10,591,556.

Therefore, the weights are 43.991% for the zero-coupon bond and 56.009% for the semi-annual bond. As a result, the weighted-average modified duration is

(4.81928 × 0.43991) + (7.71210 × 0.56009) = 6.43953.

2. Calculate the weighted-average convexity for the portfolio.

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Solution:

Using the weights calculated previously,

Weighted-average convexity = (27.87052 × 0.43991) + (72.54897 × 0.56009) = 52.89463.

3. Calculate the estimated percentage price change of the portfolio given a 100 bp increase in yield-to-maturity on each of the bonds.

Hide Solution

Solution:

$\%\Delta PV^{Full} \approx (-6.43953 \times 0.0100) + [\frac{1}{2} \times 52.89463 \times (0.0100)^2] = -6.1751\%$.

replacing the 10-year bond with?

A. A 20-year bond
B. A 15-year floating-rate bond
C. A 10-year bond with a lower coupon

Hide Solution

Solution:

B is correct. With interest rates expected to rise, the investor should choose a bond with lower interest rate risk. Both the 20-year bond and the 10-year bond with a lower coupon should have higher durations than the current 10-year bond, all else equal, and would be expected to experience a larger percentage price decrease if interest rates increase. Floating-rate bonds, however, have low interest rate risk because coupon payments adjust to changing interest rates.

OPTIONAL UNTESTED SEGMENT

UNLOCKING REAL-WORLD APPLICATIONS WITH BLOOMBERG

Bloomberg is one of the main market data platforms used globally by finance professionals throughout their workflows. In this optional section, we will explore a real-world application of the CFA Curriculum on Bloomberg.

{PORT} PORTFOLIO & RISK ANALYTICS

The Bloomberg PORT function allows you to use a portfolio optimizer to target a chosen convexity for an overall portfolio.

> "PORT <GO> empowers you to gain deeper insight into your portfolios by providing the tools to understand the structure of your portfolios, analyze your positions and active bets, and explain the drivers of historical performance and potential sources of future risk. PORT provides a streamlined workflow that includes intraday performance monitoring, fundamental characteristics, historical performance attribution, ex-ante tracking error, scenario analysis, and portfolio optimization." -Bloomberg

To explore this function, type PORT VAR into the Bloomberg command line and press Enter or <GO>. Click on the red "View" button at the top left, and select "Create New View." Type "BONDVIEW" into the "View Name" amber field, and ensure that the "New View with" amber dropdown menu reads "Equity View Settings." Then click on the grey "Create" button.

Click on the top left amber dropdown menu, select "More Sources," and choose "Funds/ETFs/13Fs." In the "<Add Fund, ETC or 13F>" amber field, insert "VUSTX" (a Vanguard bond fund), and then select it from the popup menu. VANGUARD LONG-TERM TREASURY FUND will appear on the list of names. Ensure the name is highlighted on the list. Then click on the gray "Select" button.

In the "vs" amber field, click on the dropdown menu, select "More Sources," and choose "Indices." In the "<Add Index>" amber field, insert "LUTLTRUU" (BLOOMBERG US LONG TREASURY TOTAL RETURN INDEX VALUE UNHEDGED), and then select it from the popup menu. BLOOMBERG US LONG TREASURY TOTAL RETURN INDEX VALUE UNHEDGED will appear on the list of names. Ensure the name is highlighted on the list. Then click on the gray "Select" button.

Once the portfolio loads, click on the red "Trade Simulation" button at the top of the screen and select "Launch Optimizer." Click on the gray "Add" button in the "1. Goals" section. Click into "Characteristics," then "Fixed Income," then "Risk Measures," and finally "OAS convexity (Spot)." Then click on the gray "Select" button.

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of the screen, and wait for the optimizer to run. Now you can view the trades needed to maximize the convexity of the portfolio.

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